

Duration: 3hrs

[Max Marks:80]

- (1) Question No 1 is Compulsory.
 (2) Attempt any three questions out of the remaining five.
 (3) All questions carry equal marks.
 (4) Assume suitable data, if required and state it clearly.

- 1 Attempt any **four** [20]
 a) What are Feed Forward Neural Network?
 b) Explain Gradient Descent in Deep Learning.
 c) Explain the dropout method and its advantages.
 d) What are Undercomplete Autoencoders?
 e) Explain Pooling operation in CNN.
- 2 a) What are the Three Classes of Deep Learning, explain each? [10]
 b) Explain the architecture of CNN with the help of a diagram. [10]
- 3 a) What are the different types of Gradient Descent methods, explain any three of them. [10]
 b) Explain main components of an Autoencoder and its architecture. [10]
- 4 a) Explain LSTM model, how it overcomes the limitation of RNN. [10]
 b) What are the issues faced by Vanilla GAN models? [10]
- 5 a) What are L1 and L2 regularization methods? [10]
 b) Explain any three types of Autoencoders. [10]
- 6 a) What is the significance of Activation Functions in Neural Networks, explain different types Activation functions used in NN. [10]
 b) What are Generative Adversarial Networks, comment on its applications. [10]
